We thank the reviewers for their constructive and supportive feedback. We have revised the manuscript substantially while keeping the contribution focused and within the imposed page limit. In particular, we clarified the study corpus and denominators, expanded the methodological description, modified figures, strengthened the reproducibility discussion, refined the FAIR+S framework operational scope, and added a preliminary checklist and referenced benchmarking materials. Where a reviewer requested extensive additional analysis, we provide a concise treatment in the paper and identify the corresponding work as ongoing, since a full validation would exceed the scope and page constraints of the current contribution.

Response to Reviewer 1

Comment: Some of its key percentage claims are based on a filtered subset rather than on the full review corpus, and this is not always made explicit, which creates a risk of misinterpretation.

We agree and have clarified this point throughout the manuscript. The revised abstract explicitly distinguishes the 313 publications in the underlying systematic review from the 134 publications selected for the present performance-oriented analysis. The abstract now states that the reported 89 (66.4%), 25 (18.7%), and 3 (2.2%) results refer specifically to the 134-paper subset. We also explicitly state that these 134 publications constitute the source corpus for the metric mining and taxonomy construction. This clarification avoids interpreting the percentages as representative of the full 313-publication corpus.

Comment: The paper that argues for FAIR and reproducibility does not yet provide its own coding artefacts, normalized mappings, corpus details, or agreement information, which is a serious credibility gap.

We have expanded the Metric Mining Procedure to describe the analysis as three traceable artefacts: the publication corpus and eligibility information, the extracted raw metric terms, and the normalized taxonomy and category assignments. We also state that the publication-level corpus and extracted terms will be made available through the systematic-review repository. Due to the page limit, we have kept the description concise and refer to the existing review repository rather than reproducing the complete extraction protocol in the paper.

Comment: The framework itself is also still somewhat abstract and would become much stronger if accompanied by a reporting checklist, template, or concrete worked example.

We addressed this by making the framework more operational. The revised manuscript now explicitly distinguishes computational, memory, communication, storage, and infrastructure-related components and requires the system boundary to be stated for carbon assessments. We also provide a preliminary FAIR+S checklist as an external research artefact. Since the framework and checklist are explicitly presented as work in progress, we avoid overstating their validation status. A fully validated case-study evaluation and mature reporting template are identified as subsequent work; providing a complete validation within the current paper would substantially expand the manuscript beyond its intended scope and page limit.

Comment: Claims concerning FAIR adoption and sufficient hardware context should be more carefully qualified.

We clarified the denominator and the meaning of the hardware-context result. The revised text specifies that the 25 papers are part of the 134 performance-oriented publications and describes the reported hardware information explicitly. We have also avoided presenting the result as evidence that precise carbon estimation is possible in all 25 cases. Instead, the manuscript emphasizes that missing hardware and runtime information limits reliable energy attribution.

Response to Reviewer 2

Comments: The methodology is understandable, but a slightly larger summary of the systematic literature review might help the reader to understand the background a bit more. As the systematic review is reported in a separate publication, a short overview of the selection process might also help to followe the contribution easier.

We have expanded the methodology section while retaining the detailed systematic review in its separate publication. The revised manuscript now reports the initial 2,939 records, the 494 records remaining after relevance screening, the 117 duplicates removed, and the final corpus of 313 publications, consisting of 266 articles and 47 preprints. We then explicitly describe the subsequent selection of 134 performance-oriented publications used in the present analysis. This provides sufficient methodological context while avoiding duplication of the full systematic review within the page-limited paper.

Comment: Clarify the role of preprints and publication bias.

The revised corpus description now explicitly reports the number of articles and preprints at both the initial-search and final-corpus stages. We acknowledge in the review paper that the inclusion of preprints may affect corpus composition and reporting completeness. A more extensive statistical comparison of preprints and peer-reviewed publications was not included because it would substantially extend the scope of the paper; the present study instead reports the corpus transparently and limits its claims to the defined sample.

Comment: The discussion could briefly clarify which components should ideally be included in an operational assessment, so CPU, memory, network communication, storage and/or the infrastructure overhead?

We added a dedicated paragraph defining the operational scope. The revised manuscript distinguishes computational resources (CPU/GPU), memory, network communication, storage, and infrastructure overhead such as PUE. It also emphasizes that the system boundary should be explicitly stated for each carbon assessment. This addition directly addresses the reviewer’s question while remaining concise.

Comment: Figure 2 visual presentation could be refined by reducing the empty space, also making the relation between FAIR+S and the 3 layers more explicit.

Figure 2 has been revised visually, and the framework is now presented explicitly as a three-layer structure covering performance, resource attribution/utilization, and sustainability. The surrounding discussion also clarifies how FAIR+S provides the methodological foundation for the three layers.

Response to Reviewer 3

Comment: In literature review, a discussion must be added about the possible publications bias especially when preprints are included in the study.

The revised methodology explicitly reports the inclusion of preprints and distinguishes them from peer-reviewed articles. We acknowledge in the review paper the potential effect of this choice on corpus composition and reporting practices. However, as we do not base any scientific arguments on results reported in preprints, but only identify and extract the metrics they use, this choice does not substantially affect the validity of our findings. Rather, the inclusion of preprints extends the scope of the review towards a multivocal review that incorporates grey literature, which is particularly appropriate for a metric-mining and metric-identification study. Given the limited paper length, we chose to provide a transparent methodological qualification rather than introduce a separate statistical bias analysis, which would require additional data and substantially broaden the scope of the study.

Comment: At least one case study has to be included to validate the proposed framework.

We have strengthened the practical operationalization of the framework by adding a preliminary FAIR+S checklist and explicitly defining the measurements and system components required for an operational PCN assessment. A full empirical case-study validation is not yet included because the current work is explicitly positioned as a work-in-progress framework contribution. The manuscript now makes this limitation explicit and identifies validation on a concrete PCN implementation as future work rather than implying that the current framework has already been fully validated. There is also referenced literature that represents case studies, which we can not discuss in full due to page limit.

Comment: Benchmarking of the FAIR+S with current available frameworks & Green Software Foundation guidance has to be included.

We have added benchmarking material comparing FAIR+S with currently available frameworks and the guidance of the Green Software Foundation in the appendix linked. This now complements the discussion in the main paper. Because the main contribution is the integration of reproducibility and sustainability concerns for PCN research, the detailed comparison is kept outside the main narrative to respect the page limit.

Comment: Add a frequency distribution chart/ histogram for quantified results representation.

We retained the word cloud because it provides a compact visualization of the raw terminology extracted from the literature, while the normalized frequencies are already quantitatively reported in Table 1. Importantly, the revised manuscript now explicitly states that the percentages in Table 1 represent the proportion of normalized metric mentions rather than the proportion of publications. This clarification addresses the potential ambiguity while avoiding another figure within the page-limited manuscript.

Comment: Add SCI estimation and CodeCarbon output for one PCN implementation.

The revised manuscript strengthens the methodological connection to both SCI and CodeCarbon and explains how operational energy, carbon intensity, functional units, and embodied carbon relate to the proposed framework. However, a complete new PCN implementation experiment using CodeCarbon, together with an empirically grounded SCI value, would require an additional experimental setup and validation that are beyond the scope of this short paper. We therefore present these tools and formulations as the measurement basis of the proposed framework and identify concrete implementation-level validation as ongoing work. This limitation is now stated explicitly. A related experiment is also reported in our other work (referenced in the paper); however, due to the page limit, we cannot discuss it in detail in the present paper.